

A Study on the Impact of the Occlusion Level Distribution in Datasets on the Robustness of Target Detection

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Abstract—Detection of occluded targets remains a challenge in computer vision. Collecting a large amount of occlusion datasets is costly, and current data augmentation methods for simulating occlusion do not consider the impact of occlusion level distribution. Various schemes were designed based on the diversity and the mean value of occlusion level to study the impact of occlusion level distribution in datasets on the robustness of target detection. To facilitate experiments, a novel benchmarking dataset NPOD-L (a Novel PASCAL3D+ Occlusion Dataset for Learning) with occlusion level labels is introduced to simulate occlusion with different levels. The new dataset contains samples of 10,200 images of 12 categories. By training on each scheme and subsequently testing on a test set using the YOLOv8 model, we investigated the effects of those schemes on the model's performance. The results show that there exists an optimized proportioning scheme for the occlusion degree distribution such that the mAP (mean Average Precision) of the dataset is reached in most cases. We validate the conclusions' generality through experiments on NOO-DLES1 and TOOL-D occlusion datasets, showcasing case-specific distributions' benefits and practical efficacy. This serves as a useful reference for enhancing model performance via data augmentation and regularization techniques.

Keywords—computer vision occlusion level distribution, occlusion objects detection, data enhancement

I. INTRODUCTION

Target detection is an important research direction in the field of computer vision. With continuous advancements in computer technology, especially the rise of deep learning, target detection technology has been rapidly developed and applied. Deep learning utilizes multilayer computational models to learn abstract data representations, which can discover complex structures in big data and provide powerful technical support for target detection. In recent years, target detection algorithms based on convolutional neural networks (CNNs) have achieved remarkable results [1], which not only improve the accuracy of detection, but also accelerate the detection speed, making real-time applications possible.

However, the occlusion problem has been a challenge that is difficult to avoid in practical applications of target detection. When a target is occluded by other objects, the accuracy and robustness of the target detection algorithm are seriously affected. How to effectively solve the occlusion problem and improve the performance of target detection algorithms in occlusion scenarios has become a hot topic and a difficulty.

In order to deal with the occlusion problem, researchers continue to explore new solutions. For example, in terms of algorithmic enhancement, [2] proposed a novel target detection framework for robust target detection in

occlusion situations. Another intuitive and effective one is to train the model on datasets covering different occlusion situations, [3] improves robustness by a simple regularization technique that randomly masks out square regions during training, [4] builds upon [3] by proposing a more structured information discarding enhancement to the input image, and [5] proposes to train the detector on unobscured images but allows for the learning of knowledge (e.g., semantic spatial relationships between parts, etc.). In the field of computer vision, data augmentation is a very important technique [3] to make models more robust to occlusion situations.

On the one hand, excessive occlusion in the training set may result in objects that are not informative enough to be learned for classification, contextual information is removed, and the images are more like noisy data. On the other hand, an excessive proportion of mild occlusion in the training set may cause the model to focus excessively on overall semantics and ignore partial semantics, which also brings about an overfitting probability. It is crucial to achieve an optimized ratio of information removal and retention in the occlusion dataset so that the detector is sufficiently informative about both partial semantics and overall semantics at the same time.

In order to investigate the distribution of occlusion level in the dataset, this study needs to construct a series of sample sets covering occlusion data. Given the high cost of manually labeling occlusion information for data integration, this study adopts the data-forming strategy in the literature [6], and innovatively successfully simulates a variety of occlusion scenarios by superimposing the target objects in the MS-COCO dataset onto the background of the PASCAL3D+ dataset. We created a new occlusion dataset, NPOD-L (a Novel PASCAL3D+ Occlusion Dataset for Learning), which not only contains 12 different kinds of category information and occlusion degree annotations, but also details the occlusion degree and the category of each occluded object for each sample.

We designed various occlusion level distribution schemes for the datasets, and then trained and evaluated the model's performance to identify one or more optimized occlusion level distribution schemes that could significantly enhance the robustness and accuracy of the algorithm in dealing with complex occlusion scenarios. To implement these distribution schemes, we proposed an innovative screening method to select images from the NPOD-L dataset, thereby creating subsets that meet the requirements of each distribution scheme. This method ensures that each subset maintains a consistent total number of samples while also maintaining a uniform distribution of occlusion levels across various object categories within each subset.

In order to demonstrate the effectiveness of this optimized distribution, we conducted validation experiments on the independent indoor occlusion datasets NOO-DLES1 [7] and TOOL_D, and the experimental results corroborated the validity of our conclusions.

II. RELATED WORK

A. Deep Learning Based Object Detection Techniques

Object detection algorithms are primarily divided into two categories: two-stage and one-stage. Two-stage algorithms, such as Faster R-CNN, have been extensively studied. Some research has employed the Faster R-CNN detection framework [8], utilizing a two-module object detection network that achieves higher accuracy on the KITTI dataset compared to Faster R-CNN. Although two-stage algorithms are more precise, their high computational complexity makes them unsuitable for real-time grasping tasks.

One-stage algorithms, including YOLO, SSD, ASSD, and SAPD, perform object localization and classification in a single forward pass, significantly enhancing computational efficiency while maintaining high accuracy, thus enabling real-time detection. Since the YOLO series was first introduced by Joseph Redmon et al. in 2015 [9], it has held a prominent position in the field of object detection due to its efficiency and real-time capabilities. YOLOv8 [10], building on the success of YOLOv5, incorporates innovative techniques such as AIFI (Attention-based Intra-scale Feature Interaction) and adopts a new backbone network. Its excellent training speed and accuracy provide strong performance support for our experiments.

B. Data Collection

Many existing datasets contain occluded scenes, but due to the presence of occlusion, the boundaries of the target objects may be blurred, leading to inaccurate labeling. This inaccurate labeling may affect the training of the algorithm and its performance in occluded scenes.

The data integration of manually labeled occlusion information is costly, a good one is to use synthetic datasets to simulate the occlusion situation. For example, ASMO [11] proposed by Li et al. can automatically merge objects using different distances and angles between them, this makes the synthetic dataset balanced in terms of the degree of occlusion and occlusion, and the dataset generated by ASMO covers a more diverse range of occlusion scenarios compared to the one that simply randomly superimposes an object onto the occluded object, thus demonstrating a better performance in the comparison. However [11] mainly focuses on comparisons with pre-synthesis and randomized synthetic datasets without further delving into how changes in the occlusion degree distribution affect the robustness of the algorithm, which leaves room for exploration in subsequent research.

Wu et al. created the NOO-DLES1 [7] indoor occlusion dataset to explore the effects of occlusion degree, occlusion position, and occluded position on the model, but since the five research objects in the dataset are indoor objects with relatively simple characteristics such as plates and cups, and the data volume of each leaf node

dataset is tens of sheets, this limitation of the data volume may restrict the results to a certain extent of generalization and depth.

In order to overcome these problems, two public and large-scale datasets, PASCAL3D+ and COCO, are used in this study. These datasets not only have more sufficient data volume, but also cover a wide range of research objects, thus providing richer and more comprehensive data support and higher reference value for the subsequent applied research.

C. Data Enhancement

Some of the information erasing ones proposed in data enhancement have also achieved results, such as random Erasing [12], Hide-and-Seek [13], Cutout [3], GridMask [4], which counteract the possible overfitting problem of CNNs by deleting some pixel points on the input image and introducing noise, but too much information erasing may cause the image to lose too much critical information. Avoiding information erasure and retention is the core requirement for making occlusion datasets.

Cutout investigates the relationship between the size of the excision patch and the validation accuracy, Random Erasing compares the performance of models when using different padding values, and GridMask explores the correlation between the spacing of the excision grid and the validation accuracy. Since the size and location of the target object in each image are different, there exists the possibility of deleting only to the background information or completely deleting the target information, it is difficult to uniformly determine an information deletion ratio that applies to all samples.

From the above discussion, it can be observed that although some of the proposed data enhancement methods were able to achieve better detection accuracy for occluded objects, there is still room for improvement. Therefore, this paper delves into this issue in depth, providing a new reference point for future, more effective data enhancement strategies.

III. DATASET AND BASELINE

A. NPOD-L dataset construction

Due to the high cost of manually annotating occlusion information in data integration, this study adopted open-source dataset synthesis code [6]. The PASCAL3D+ dataset has a large amount of data, multiple categories, and covers segmentation mask information for the samples. By overlaying target objects from the MS-COCO dataset with three occlusion levels (mild occlusion: 10-30%, moderate occlusion: 30-50%, severe occlusion: 50-70%) onto the original unobstructed images in the PASCAL3D+ dataset, we successfully simulated different levels of occlusion scenarios, including 12 types: airplane, bicycle, bus, car, motorbike, train, boat, bottle, chair, dining table, sofa, and TV monitor. For each unobstructed image, three composite images of different occlusion levels were generated. There are 10,200 images for each occlusion level, totaling 30,600 images, as shown in the figure. Figures 1(a), (b), (c), and (d) provide examples of several types of objects that are unobstructed, lightly occluded, moderately occluded, and heavily occluded.

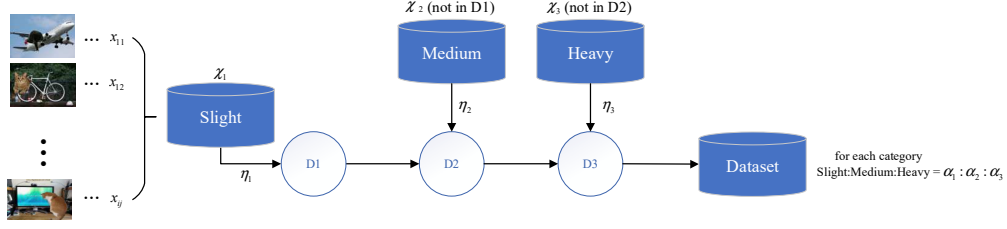


Figure 1. Collection process of sub dataset.

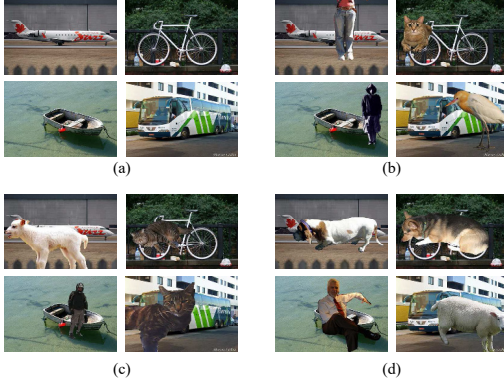


Figure 2. Examples of different levels of shading.

B. Sub-dataset Selection

To ensure consistency in the number of samples for each training model, we randomly split the data for each occlusion level (e.g., mild, moderate, severe) into a 6:2:2 ratio, resulting in a training set of 6000 images, a validation set of 2000 images, and a testing set of 2000 images for each occlusion level. The collection standards are as follows:

1) *Uniform Total Data Amount*: The total amount of data allocated to each occlusion level for training, validation, and testing is the same, ensuring fairness across levels.

2) *Avoidance of Duplicate Images*: Avoid the same base image appearing multiple times with different levels of occlusion.

3) *Consistent Occlusion Distribution*: The occlusion distribution of each category is consistent with the overall occlusion distribution.

Firstly, the total data volume is set to N . The ratio of sample sizes for slight, medium, and heavy levels of occlusion is $\alpha_1 : \alpha_2 : \alpha_3$. Samples are first classified according to the degree of occlusion and the target category, and then proportionally extracted from each category in order to ensure that the occlusion proportion of each category is consistent with the overall proportion. The required number of images for each category is n_{ij} (1):

$$\eta_i(n_{i1}, n_{i2}, \dots, n_{ik}) = \frac{\alpha_i N}{\sum_{i=1}^3 \alpha_i \sum_j x_j} \chi_i(x_{i1}, x_{i2}, \dots, x_{ik}) \quad (1)$$

Where η_1, η_2, η_3 is the vector of required class samples under slight, medium, and heavy occlusion levels,

TABLE I. PROPORTION RATIO SCHEME

<i>Scheme</i>	<i>Proportion</i>	<i>Diversity</i>
A1	1:0:0	Low
A2	0:1:0	
A3	0:0:1	
B1	1:1:0	Medium
B2	1:0:1	
B3	0:1:1	
C1	1:1:1	Hight
C2	1:2:1	
Baseline	the Dataset without Occlusion	

k is the number of classes. χ_i is the vector of class samples under slight, medium, and heavy occlusion levels in the source dataset.

Since all three occlusion datasets originate from the same non-occluded dataset, random selection within each occlusion level may cause the duplication of the same base image with different levels of occlusion. According to the plan outlined in (1), the process will involve first selecting images from the mildly occluded set as D1, then adding non-homologous images from the moderately occluded set as D2, and similarly for the severely occluded images, to form the final dataset. This workflow is illustrated in Fig. 2.

C. Different Distribution Schemes

Based on the diversity of occlusion levels and the average occlusion level, various distribution schemes have been designed to simulate different occlusion scenarios. The proportions indicate the data volumes for slight, medium, and heavy occlusions, as shown in Table I.

Parts A and B specifically aim to investigate the impact of average occlusion rates on the robustness of the models under both single occlusion and mixed occlusion distributions. In these sections, we control the diversity of occlusion levels and adjust the average occlusion level to evaluate how these factors influence model performance. On the other hand, Part C focuses on assessing the effect of occlusion diversity on robustness, exploring how variations in occlusion levels affect the models' accuracy in detection tasks.

The baseline refers to the source dataset that has not undergone any occlusion processing.

D. Experimental Environment

Each subset utilizes 8000 images, comprising 6000 for training and 2000 for validation. The test set is made up of 6000 pre-assigned images from NPOD-L. The test set is also divided into three levels based on the degree of occlusion: slight, medium and heavy.

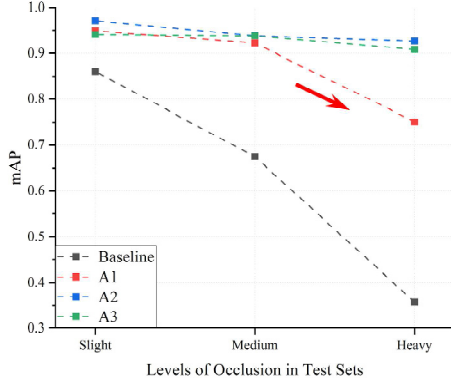


Figure 3. mAP of models trained on each single level of occlusion.

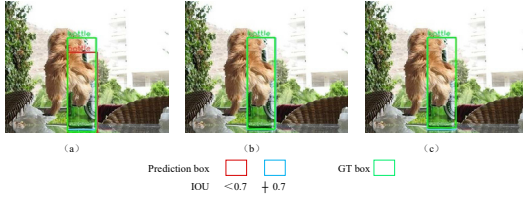


Figure 4. IoU at different occlusion levels.

The detection framework for model training is Yolov8, and the models were all trained over 100 epochs, utilizing a batch size of 16. The computer system is Windows 11, equipped with a Core i9 CPU and an NVIDIA GeForce RTX 4060 (8GB) GPU. The deep learning framework used is PyTorch.

Evaluating Indicator

We use mAP (mean Average Precision) on the test set as the evaluation metric, considering detections with an IoU (Intersection over Union) ≥ 0.7 as true positives. Each category is evaluated independently.

E. Performance under each Single Degree of Occlusion

In order to find the optimization parameters for the ratio, we first independently train for four levels of occlusion: no occlusion, mild occlusion, moderate occlusion, and severe occlusion. Each training set only contains a single occlusion ratio.

Fig.1 showcases the mAP of various schemes on test sets with differing occlusion levels.

From the perspective of the occlusion level of the test machine, severe occlusion will cover key target features, resulting in a sharp decrease in accuracy.

Compared with the Baseline, each scheme is better than the baseline (non-occluded training), especially in the case of mild and moderate occlusions.

In terms of the performance of each solution, under heavy occlusion, Scheme A1's mAP significantly drops, while Scheme A2 and Scheme A3 maintains high mAP. Among the schemes, Scheme A2 exhibits the best mAP of 92.66%. Compared to the baseline, Scheme s A1 and A3 show increases in mAP by 52.92%, 17.56%, and 1.85%, respectively.

This performance matches our expectations. Schemes A2 and A3, with higher avg. occlusion, enable the detector

TABLE II. MAP UNDER DIFFERENT LEVELS OF DIVERSITY

Scheme	D	mAP(%)
A2	0.000	92.66
C1	0.500	91.50
C2	0.625	90.53

to focus on unoccluded semantics, improving its ability to recognize occluded objects using remaining visible features. Moreover, due to the dataset's realistic detection boxes (non-fuzzy boundaries), the detector can better learn and predict occluded parts' boxes based on context, reducing IoU insufficiency, as shown in Fig. 4(a).

(a), (b), and (c) represent the detection images from models trained on purely slight, medium, and heavy occlusion training sets, respectively. It can be observed that the detection boxes and ground truth boxes of the latter two are almost coincident.

F. Impact of Average Occlusion Rate and Diversity of Occlusion Levels

In real-world environments, various occlusion scenarios exist. This section explores the impact of average occlusion rate and diversity of occlusion degrees on model robustness. To investigate this, we introduce the Simpson's Diversity Index D to quantify the diversity of occlusion distributions, as formulated in (2):

$$D = 1 - \left(\frac{\sum n(n-1)}{N(N-1)} \right) \quad (2)$$

where n refers to the number of samples covered by each degree of occlusion, and N refers to the total number of samples.

For the average occlusion rate $\bar{\rho}$, the formula is (3):

$$\bar{\rho} = \frac{\sum \rho n}{N} \quad (3)$$

where ρ is occlusion rate, n refers to the number of samples covered by each degree of occlusion, and N refers to the total number of samples.

We controlled $\bar{\rho}$ to investigate the diversity of occlusion degrees. Based on the initial proportion of Scheme A2 (which only contained moderately occluded images), we gradually increased the diversity of occlusion levels. Initially, we attempted to evenly distribute half of the medium occlusion dataset to the slight set and the heavy set (i.e., Scheme C1), but observed a decrease mAP. Further adjustments were made until the occlusion degrees were uniformly distributed (i.e., Scheme C2), and observed the robustness of the model continued to decrease, as shown in Table II.

The results indicate that, increasing the diversity of occlusion rate distribution actually leads to a decrease in performance. This differs from the findings in [7], possibly due to the smaller scale and simpler target features of the indoor occlusion dataset used in [7], whereas our experimental dataset is larger and features more complex backgrounds. The diversity in occlusion degrees may require more parameters and a more complex architecture, which could potentially lead to overfitting and a decrease in generalization ability.

TABLE III. MAP UNDER DIFFERENT SCHEMES IN OTHER DATASETS

	AP(%)					mAP(%)
	Scheme	bottle	bowl	cup	plate	
NOO-DLES1	B1	94.20	86.43	90.20	83.64	88.62
	B2	99.69	99.77	96.82	92.94	97.30
	B3	98.30	97.21	94.22	97.14	96.72
	Scheme	screwdriver	tape	solidgum	-	
TOOL-D	B1	46.44	89.28	99.05	-	78.26
	B2	44.19	98.99	99.03	-	80.74
	B3	36.24	94.00	100.00	-	76.75
	Scheme	screwdriver	tape	solidgum	-	

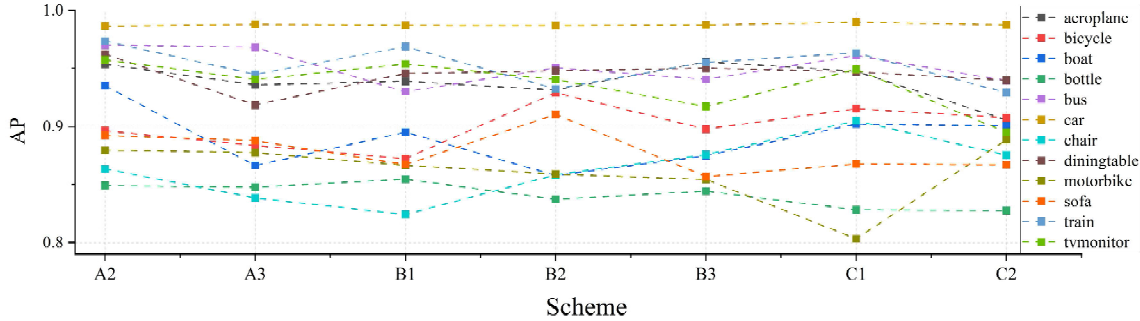


Figure 5. AP of each scheme with different types of objects.

Next, we controlled the distribution diversity and split the study into Groups A and B to investigate the average occlusion rate. Through a comparison of mAP across various scheme (as shown in Table IV), A2 performed best in Group A, and B2 performed best in Group B.

TABLE IV. MAP UNDER DIFFERENT AVERAGE OCCLUSION RATES

D	ρ	Scheme	mAP
0	0.2	A1	75.10
	0.4	A2	92.66
	0.6	A3	90.81
0.5	0.3	B1	90.87
	0.4	B2	91.18
	0.5	B3	90.92

In complex data environments, especially when the dataset is large and the feature distribution is diverse, the diversity of occlusion distribution in the model may increase the risk of instability during training. For example, problems such as gradient vanishing or gradient explosion may occur more easily, causing the training to fail to converge or converge to an undesirable solution. When the diversity of occlusion degrees is fixed, the closer the average occlusion rate is to medium occlusion (around 40%), the better the performance. Therefore, when designing and preparing training data, reasonably controlling the diversity of occlusion degrees and ensuring that the average occlusion rate is close to the intermediate value is an effective strategy for improving model performance and generalization ability.

G. Impact of Each Scheme on Different Types of Objects

Let's look at each type separately. Fig.5 shows the line chart of the AP of different types of objects for training

sets with different ratios. Although there is no single solution that will optimize performance in every category, A2 and B2 are more stable.

H. Validate the Conclusions' Generality on NOO-DLES1 and TOOL-D

We consider that the conclusions obtained by PASCAL3D+ may be biased towards large and complex datasets, so we introduce two small indoor occlusion datasets, NOO-DLES1 and TOOL-D, to verify the validity of the previous conclusions.

We strictly followed the standards set out in III-B to ensure the consistency and reliability of the data.

Due to the limitation of the total amount of occlusion in each of these two datasets, we adopted a diversity distribution strategy of $D = 0.5$ to maximize the utilization of the datasets. For the NOO-DLES1 dataset (as shown in Fig.6 (a)), we created a training set containing 800 images and a sub-test set containing 292 images; for the TOOL-D dataset (as shown in Fig.6. (b)), we created a training set containing 776 images and a sub-test set containing 300 images. The experimental results are shown in Table IV.

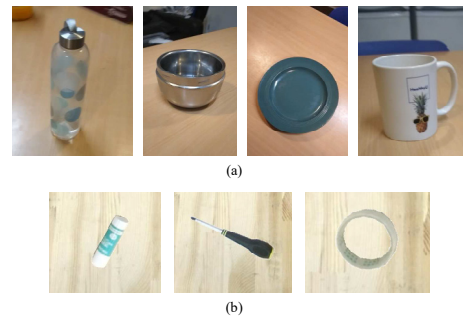


Figure 6. Example images of dataset sample.

In simple datasets with a small amount of data, according to the results of Scheme B1 and B3, for simple-structured and monochromatic items such as solid glue and plates, a higher degree of occlusion may be more conducive to the model learning how to deal with occlusion situations. For items with more features, such as screwdrivers, the occlusion degree distribution may need to take into account the needs of feature learning more, and reduce the proportion of occluded samples, so that the model can focus more on learning the features of the item.

Although other distributions perform well in specific situations, the B2 distribution still dominates in terms of overall mAP value comparison. This shows that, from the perspective of a wider dataset and general application scenarios, B2 is more likely to be a safe choice for balancing feature learning and occlusion processing. Therefore, when selecting the occlusion distribution of the training set, it is necessary to comprehensively consider the characteristics of the dataset, the complexity of the items, and the occlusion in the actual application to find the optimal training strategy.

IV. CONCLUSION

In this paper, we systematically study how to improve the robustness of the YOLOv8 object detection algorithm in severely occluded environments by optimizing the occlusion level distribution in the training data. To this end, we have constructed a synthetic dataset NPOD-L containing different levels of occlusion, which successfully simulates multi-level occlusion scenarios.

We then experimented with generalizing the results to two datasets with smaller data volumes and sample sizes: the independent indoor occlusion datasets NOO-DLES1 and TOOL-D.

The conclusions are as follows:

Complex occlusion dataset combined in dataset NPOD-L:

(1) Comparing schemes A2, C1, and C2, it can be seen that the diversity of occlusion levels is not as good as we thought. In complex datasets, the increasing of diversity of occlusion distributions may increase the risk of instability during training.

(2) Comparing groups A and B, it can be seen that the model performs best when the average occlusion rate is close to moderate occlusion (around 40%).

(3) Comparing different categories of objects, A2 and B2 are more stable (with fewer cases of mAP deterioration) and can improve overall performance, as shown by the comparison of mAP values.

Extend to simply masking the dataset to form the dataset NOO-DLES1 and TOOL-D:

(1) For items with simple and repetitive features, appropriately increasing the degree of occlusion can make the model focus on learning the recognition of objects with occlusions. Conversely, for items with more features, appropriately reducing the degree of occlusion allows the model to focus more on learning the key features of the item.

(2) Although other occlusion distributions may perform better in specific cases, from the perspective of overall mAP value comparison, the distribution of occlusion rate that is closer to the mid-range solution B2 is more robust in balancing feature learning and occlusion processing.

In summary, when selecting the occlusion distribution of the training set, it is necessary to comprehensively consider the characteristics of the dataset, the complexity of the items, and the occlusion conditions in practical applications. However, in most cases, using a moderately occluded training set or using both a lightly and heavily occluded training set may be more appropriate in terms of balancing feature learning and occlusion processing.

This paper effectively improves the robustness of the object detection algorithm in severely occluded environments by optimizing the occlusion level distribution in the training data. This finding is of great significance for promoting the deployment of object detection technology in more practical application scenarios. Future work will further explore the impact of more complex occlusion situations and larger-scale datasets on model performance, and study more advanced data augmentation strategies to improve model generalization.

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